# Earth System

# # Earth System Science

## Chapter 2: THE COMPONENTS OF THE EARTH SYSTEM

## Objective

## Biosphere and Anthroposphere

- *To Define The Ecology And Ecosystem Concept*
- *To Know The Ecosystem System Structure*
- *To see Human Caused Changes In Earth's Ecosystem*
- *To be familiarize with Anthroposphere*

# ***Biosphere***

- Also known as the **ecosphere**.
- Is the global sum of all ecosystems. It includes all regions of Earth where life exists, such as land, water, and the atmosphere.
- It's a physical layer of Earth encompassing all living organisms and their habitats.
- It is the regions of Earth where life **does** exist, overlapping with parts of the geosphere, hydrosphere, and atmosphere.

# ***What Is Ecology?***

- Ecology is the scientific study of interactions among organisms and between organisms and their environment.
- It focuses on relationships and processes, such as food chains, nutrient cycles, and energy flows.
- An ecosystem is a structural and functional unit of ecology where the living organisms interact with each other and the surrounding environment.
- Ecosystem also the integrated study of living (biotic) and non-living (abiotic) components of ecosystems and their interactions within an ecosystem framework.

# *The Ecosystem System Structure*

# Ecology

*Ecosystem*

## biotic

Producers
(Autotrophs)

Consumers
(Heterotrophs)

Decomposers
(saprotrophs)

## Abiotic

**air**

**water**

**soil**

**minerals**

**sunlight**

- The structure of an ecosystem is characterised by the organisation of both biotic and abiotic components.
- This includes the distribution of energy in our environment.
- It also includes the climatic conditions prevailing in that particular environment.
- The structure of an ecosystem can be split into two main components, namely:
  - Biotic Components
  - Abiotic Components
- The biotic and abiotic components are interrelated in an ecosystem.
- It is an open system where the energy and components can flow throughout the boundaries.

## ***Biotic Components***

- Biotic components refer to all living components in an ecosystem.
- Based on nutrition, biotic components can be categorised into autotrophs, heterotrophs and saprotrophs (or decomposers).

## ***Abiotic Components***

Abiotic components are the non-living component of an ecosystem. It includes air, water, soil, minerals, sunlight, temperature, nutrients, wind, altitude, turbidity, etc.

# *Functions of Ecosystem*

- It regulates the essential ecological processes, supports life systems and renders stability.
- It is also responsible for the cycling of nutrients between biotic and abiotic components.
- It maintains a balance among the various trophic levels in the ecosystem.

# *Interactions with the Atmosphere, Hydrosphere, and Lithosphere*

- ***Atmosphere:*** The biosphere interacts with the atmosphere through processes such as *photosynthesis and respiration*. Plants, for example, take in carbon dioxide and release oxygen during photosynthesis, influencing atmospheric composition.
- ***Hydrosphere:*** Aquatic ecosystems, including oceans, lakes, and rivers, are integral parts of the biosphere. Organisms within these systems contribute to *nutrient cycling* and play a role in regulating water quality.
- ***Lithosphere:*** The biosphere interacts with the Earth's crust through processes like *nutrient cycling and the decomposition of organic matter*. Soil, a component of the lithosphere, provides a habitat for many organisms.

- The biosphere is involved in various biogeochemical cycles, such as the carbon cycle, nitrogen cycle, and water cycle.
- These cycles involve the movement of essential elements and compounds between living organisms, the atmosphere, soil, and water

# *Climate Regulation*

- The biosphere plays a crucial role in climate regulation. Forests, for instance, act as carbon sinks, helping to mitigate the impact of greenhouse gas emissions on climate change.

# *Human Caused Changes In Earth's Ecosystem*

- People play an important part in maintaining the flow of energy in the biosphere.
- Sometimes, however, people disrupt the flow.
- For example, in the atmosphere, oxygen levels decrease and carbon dioxide levels increase when people clear forests or burn fossil fuels such as coal and oil. Oil spills and industrial wastes threaten life in the hydrosphere.
- The future of the biosphere will depend on how people interact with other living things within the zone of life.

# ***Human Impact and Conservation***

- Human activities significantly impact the biosphere, leading to habitat destruction, pollution, and climate change.
- Conservation efforts aim to protect biodiversity and maintain the health of ecosystems for future generations.

# *The Anthroposphere*

# *The Anthroposphere The Human Sphere*

- The anthroposphere (also called the *technosphere*) is the part of the Earth that has been created, modified, or influenced by human activities.
- It includes all human-made environments, systems, and their effects on the natural world.

# *Major Components*

- Urban areas (buildings, transport networks)
- Industrial systems (factories, energy plants)
- Agricultural land (cropland, irrigation systems)
- Waste and pollution (solid waste, greenhouse gases)
- Information systems (communication, data networks)

# *Why Consider It a Separate Sphere?*

- ❑ Humans now dominate global environmental change
- ❑ Alters natural cycles (carbon, nitrogen, water)
- ❑ Helps scientists study human Earth system feedbacks
- ❑ Highlights human responsibility for sustainability

# Anthroposphere – HUMAN BEINGS

The anthroposphere is anything brought about by humans that modifies the environment for activities or habitation.

- The anthroposphere encompasses the total human presence throughout the Earth system including our *culture*, *technology*, *built environment*, and *associated activities.*

BIOSPHERE
ATMOSPHERE
ANTHROPOSPHERE
LITHOSPHERE
HYDROSPHERE

….END

Thanks !